

Total no. of Pages: 1

Roll no.

Degree: B-Tech.

Semester 3rd

MID-SEMESTER EXAMINATION, June, 2026

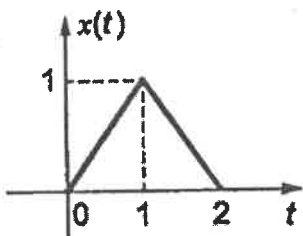
Course Title: Signals and Systems

Course Code: Sem: ECECC302/EIECC302/VTECC302

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Define a continuous-time signal and a discrete-time signal with one example each.	2	C01
1b	Define the unit impulse signal and state any two properties of the impulse function.	2	C01
2a	Determine whether or not each of the following systems are time variant with input $x(t)$ or $x[n]$ and output $y(t)$ or $y[n]$. (i) $y[n] = x[n] - x[n - 1]$ (ii) $y[n] = x[t - 1] - x(1 - t)$	2	C01
2b	Draw the even and odd components of the waveform of a signal $x(t)$ shown in figure. 	2	C01
3a	Determine the output of an LTI system if $x(t) = \delta(t)$ and $h(t) = e^{-t}u(t)$	2	C04
3b	Define a Linear System. State the principle of superposition.	2	C01
4a	State and prove distributive property of continuous time convolution	2	C02
4b	The unit impulse response of a system is given as $c(t) = -4e^{-t} + 6e^{-2t}$. Find the step response of the system for $t \geq 0$.	2	C04
5a	What is meant by orthogonality of signals? Give one example.	2	C02
5b	For the input $x(t) = \delta(t - 1) + \delta(t + 1)$ and impulse response $h(t) = u(t + 1) - u(t - 2)$, draw the waveform corresponding to output $y(t)$.	2	C02